



NUTRITIONAL ASSESSMENT (HuNu3081)

CHAPTER THREE

ANTHROPOMETRIC ASSESSMENT METHODS

Questions

1. What anthropometric measurements you know?
2. Which instruments are used to conduct anthropometric measurements ?
3. What anthropometric indices you know?

Objectives

- **At the end of this Chapter you will be able to:**
 - Identify anthropometric measurements of body size/ growth.
 - Identify anthropometric measurements used to assess body composition/ physical dimension.
 - Conduct different anthropometric measurements.
 - Construct anthropometric indices & indicators.
 - Interpret the anthropometric reference data.

Introduction

- Anthropometric measurements are the most commonly used methods for the assessment of two of the most widespread nutrition problems in the world:
 - 1) Protein-Energy Malnutrition [PEM]: (among infant & young child, pregnant & lactating women);
 - 2) Under & Over nutrition in all age-groups.
- Anthropometric measurements could be used both in the clinical and field set-ups.

Introduction...(1)

- Anthropometry comes from two Greek words: Anthropos = Human, and metry/ metron = measurement.
- Involves the measurements of the variations of physical dimensions & gross composition of the human body at different age, levels & degrees of nutrition.
- The measured values of anthropometry reflects **current nutritional status** but don't differentiate between **acute & chronic changes**.

Introduction...(2)

- Anthropometric measurements assess:
 - **Growth** (in children): weight, height/ length & head circumference.
 - **Body composition** (in adults): body fat & fat-free mass.
 - ✓ Body fat: Skin fold thicknesses, hip & waist circumferences.
 - ✓ Fat-free mass: **Mid-Upper Arm Circumference (MUAC)**.
- **Weight gain** is the most accurate and sensitive method of growth measurement.

1. Anthropometric Assessment of Body Size (Growth)

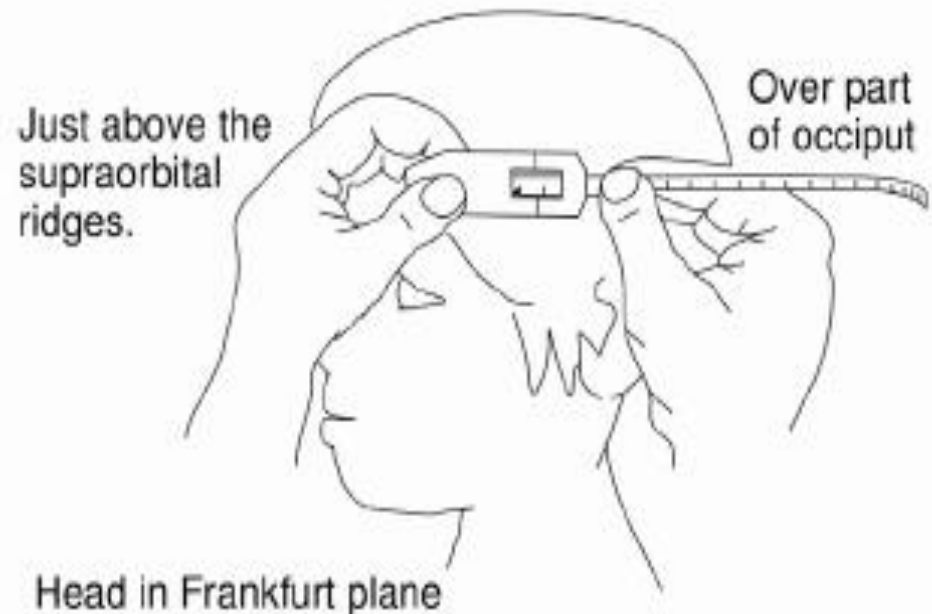
➤ Head circumference, stature (height/length) & body weight.

A. Head Circumference (HC): closely related to brain size but not necessarily related to intelligence.

- It is the circumference of the head along the supra orbital ridge anteriorly and occipital prominence posteriorly.
- It can detect pathological conditions: unusually large (**macro cephalic**) or small (**micro cephalic**) head.
- HC is useful in assessing **chronic nutritional problems** in under two children; *e.g.* HC-for age: index of chronic protein-energy-malnutrition for children < 2 years.
- But after 2 years it is not useful as the growth of the brain becomes sluggish.

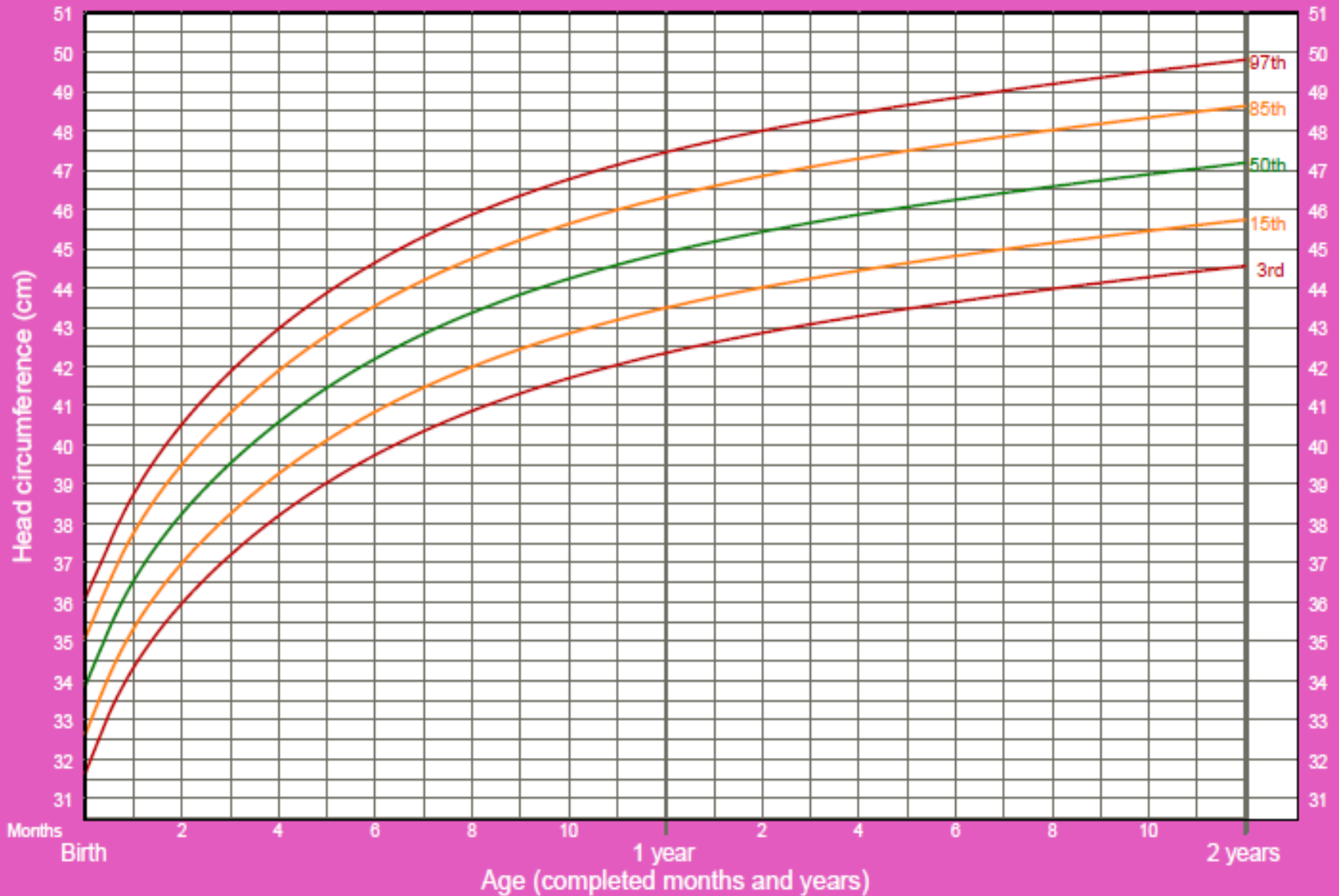
Head Circumference... (1)

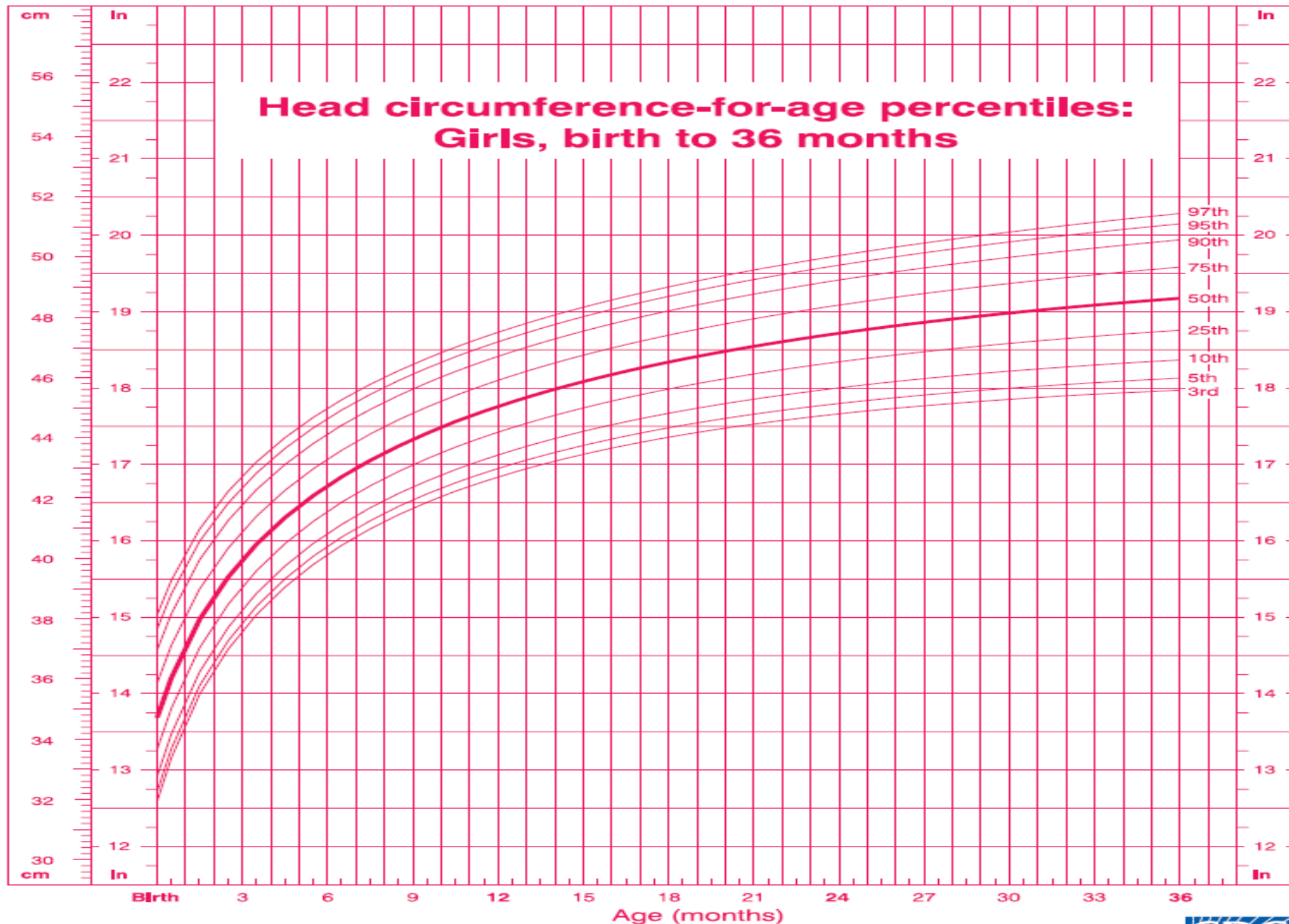
- Measured using **flexible, non stretch measuring tape**.
- Subject stand with left side facing; arms relaxed & legs apart.
- Position the tape just above the eyebrows, above the ears & around the biggest part on the back of the head.
- Ensure tape at same level on each side of head; pull tightly to compress hair & measure to nearest millimeter.



Head circumference-for-age GIRLS

Birth to 2 years (percentiles)





Published May 30, 2000.

SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).

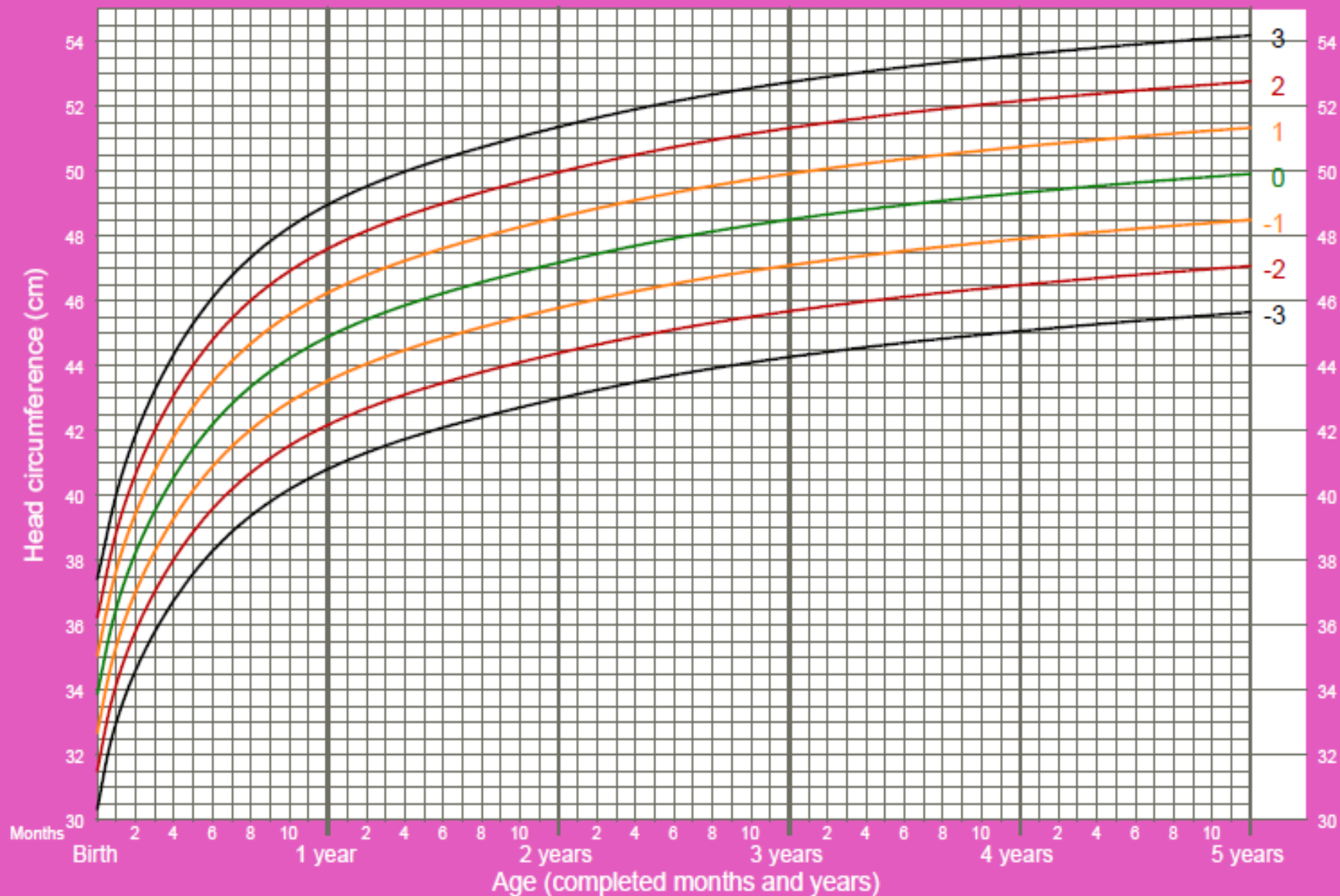


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Figure 8. Individual growth chart 3rd, 5th, 10th, 25th, 50th, 75th, 90th, 95th, 97th percentiles, birth to 36 months: Girls head circumference for-age

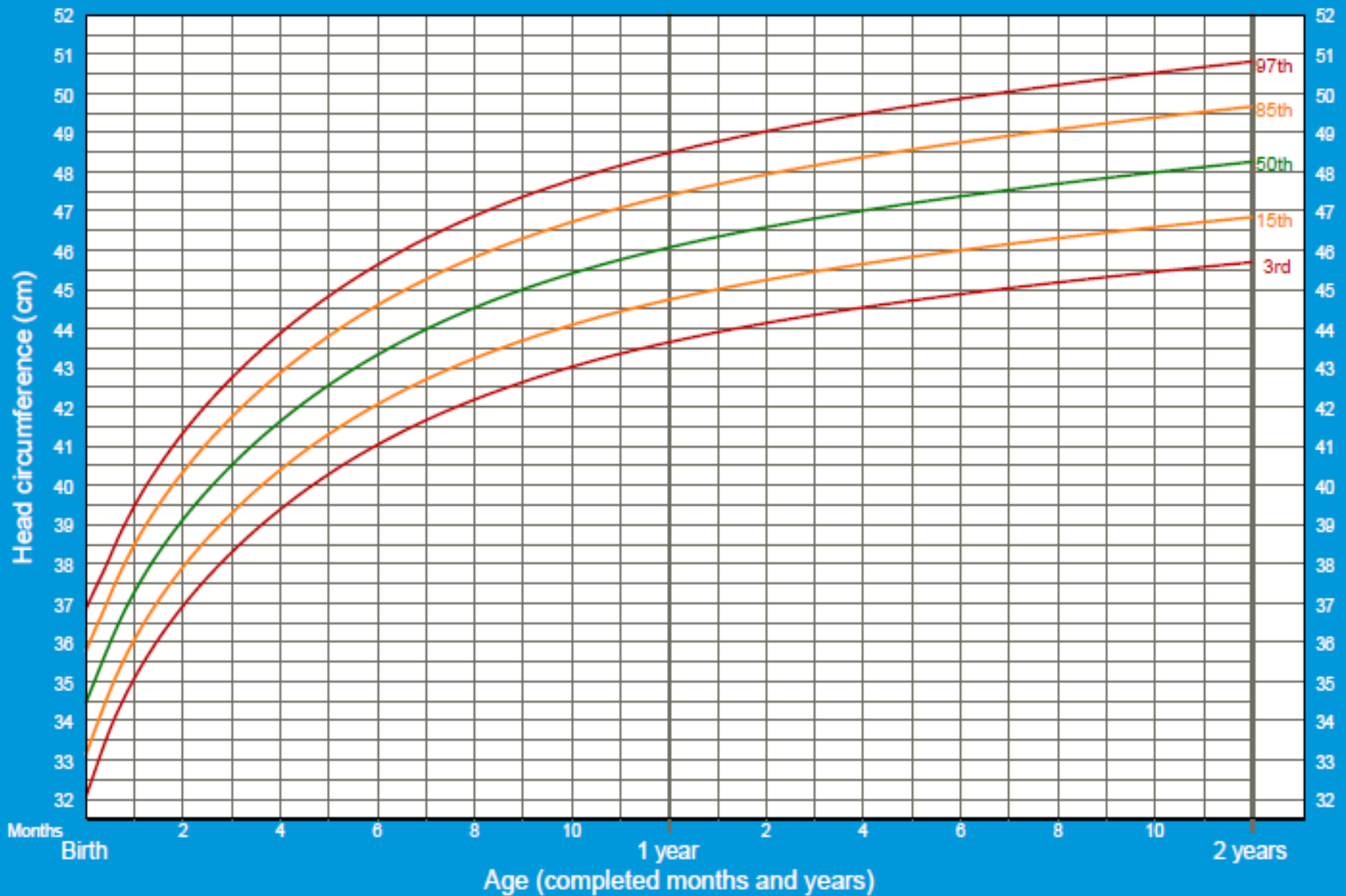
Head circumference-for-age GIRLS

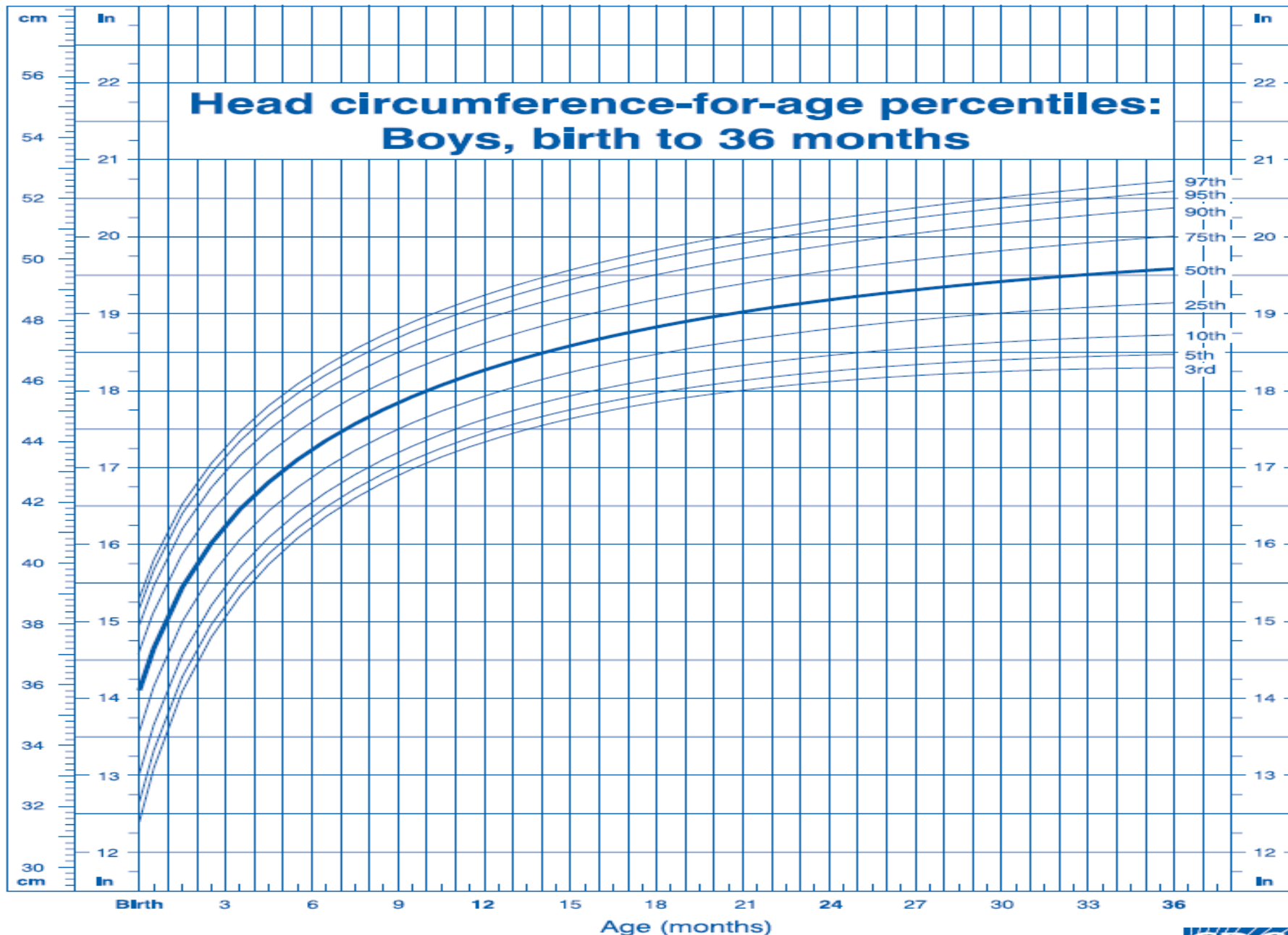
Birth to 5 years (z-scores)



Head circumference-for-age BOYS

Birth to 2 years (percentiles)





Published May 30, 2000.

SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).

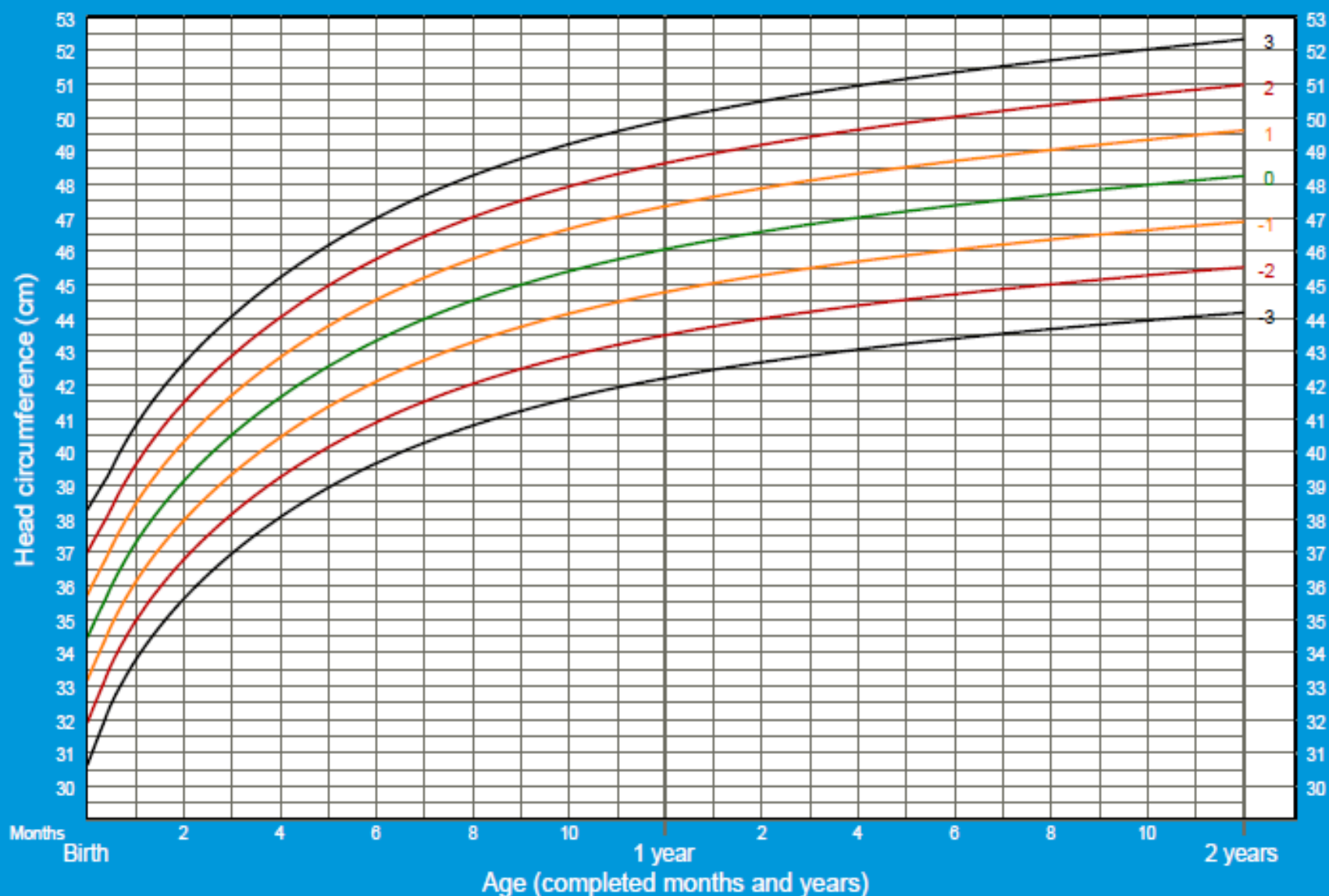


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Figure 7. Individual growth chart 3rd, 5th, 10th, 25th, 50th, 75th, 90th, 95th, 97th percentiles, birth to 36 months: Boys head circumference-for-age

Head circumference-for-age BOYS

Birth to 2 years (z-scores)

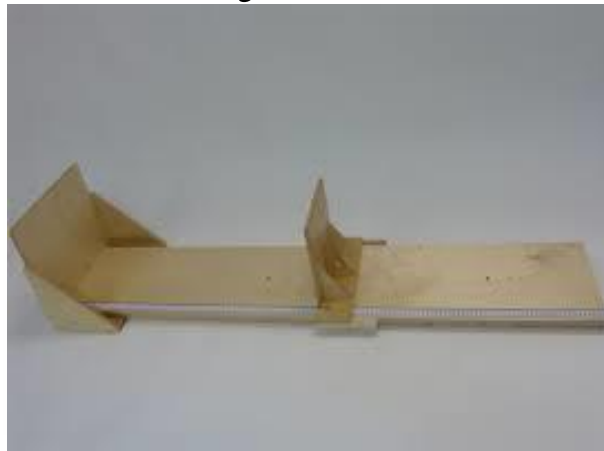


Anthropometric Assessment of Body Size (Growth)... (1)

B. Recumbent Length (*Lying back / horizontal*)

- Used for infants & children <2yrs & having height $\leq 85\text{cm}$.
- Measured using wooden measuring board/ recumbent length board or sliding board
- Measurement is read to the nearest millimeter.
- Two examiners are required to correctly position the subject & ensure accurate & reliable measurement.
 - One examiner holds the child's head in a frankfurt position.
 - The other examiner holds the subject's feet.

Recumbent length boards



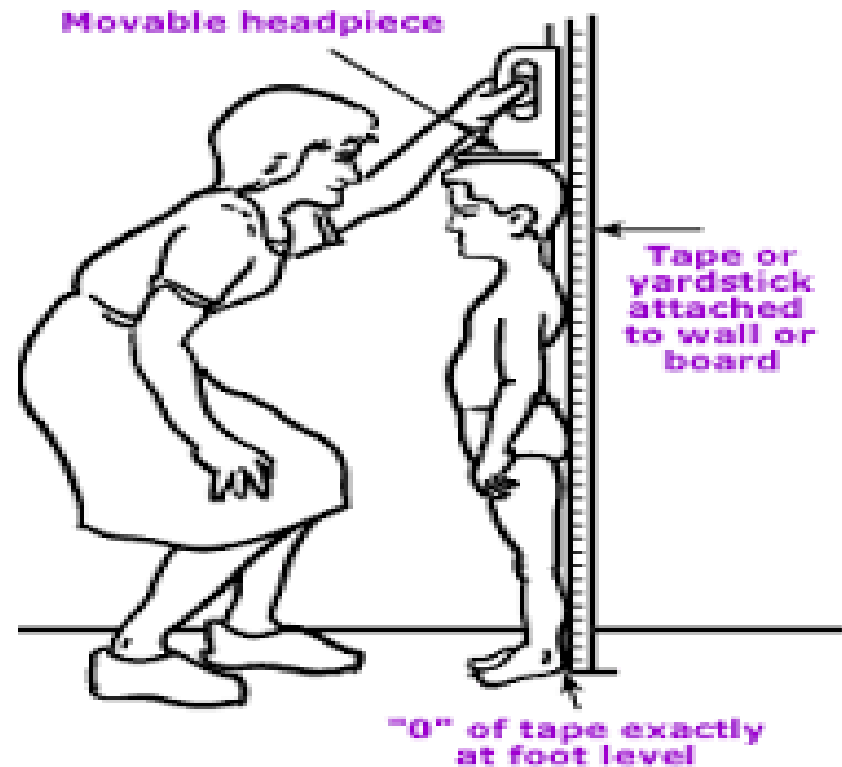
Measurement of recumbent length



Anthropometric Assessment of Body Size (Growth) ... (2)

C. Height

- For children above 2 years of age, height $>85\text{cm}$ & for adults.
- Measured in the standing position using a stadiometer.
- There should be minimal clothing to see posture clearly.
- Height is recorded to the nearest 0.5 cm.



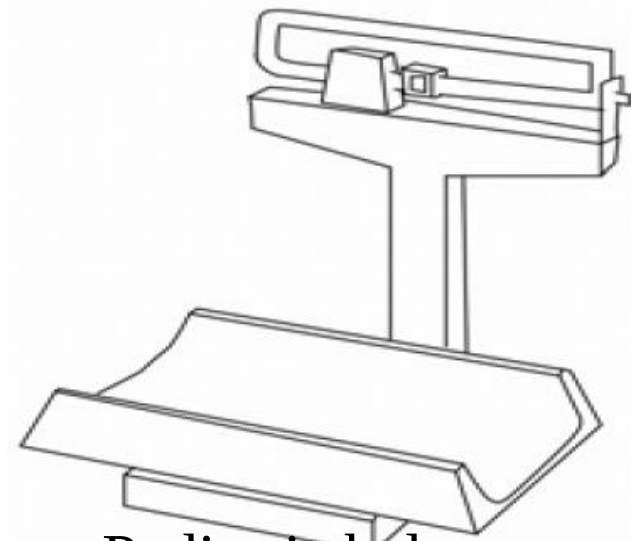
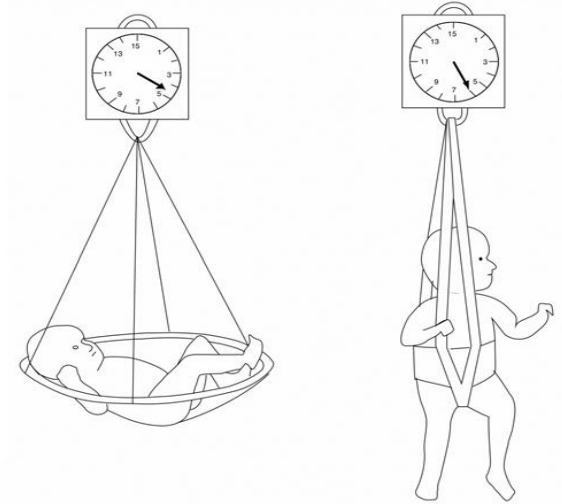
Anthropometric Assessment of Body Size... (3)

D. Weight in Infants & Children

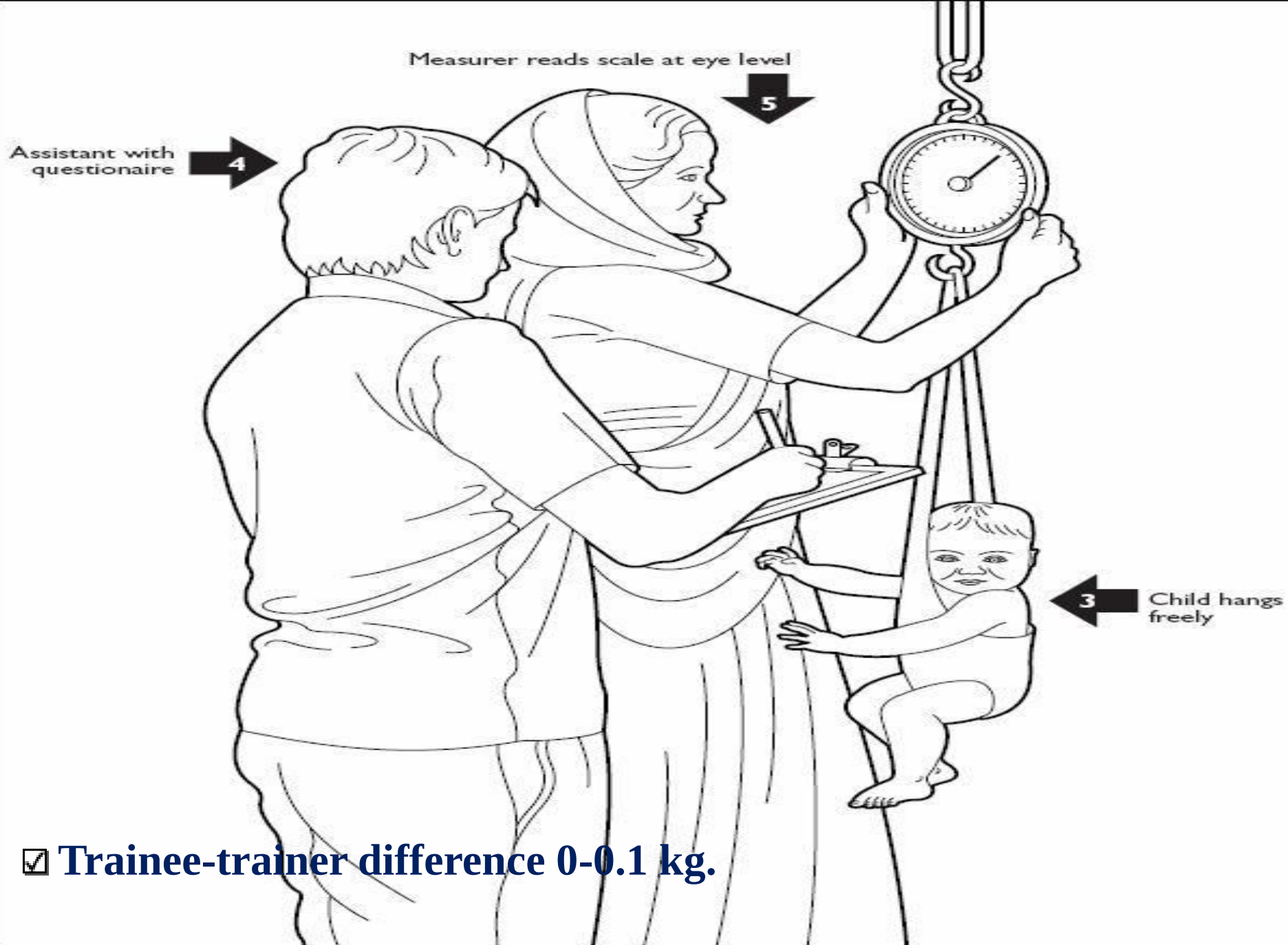
- Suspended scale, weighing sling (spring balance) & pediatric balance can be used for infants & children ≤ 2 years of age.
- In children the measurement is performed to the nearest 10g.
- In adults and children >2 years, beam balance can be used and the measurement is performed to the nearest 0.1 kg (100g).
 - Digital (electronic) scales can be used for both adults & children and are very accurate.
- To measure a child's weight (without digital (electronic) scales):
 - Weigh subjects together with the mother (A).
 - Weigh the mother alone (B).
 - Subject's weight = A-B.

Measurement of weight in infants & children

Suspended/ spring scales for those < 2 years (less reliable).



Pediatric balance





E. Weight in older Children & Adults

- Preferably done after the bladder is empty or before meal.
- The balance should be placed on hard & flat surface, adjusted for zero balance.
- Subject stands at center with minimal clothing, look straight ahead, relaxed & stand unassisted.
- Balance should be calibrated with a set of weights.



Seca



Beam balance

Anthropometric Indices: Growth

- Derived from two or more *raw measurements*.

1. **Weight-for-Height or Length (WFH/L):**

- A sensitive index for the assessment of current nutritional status (acute malnutrition).
- ✚ Low weight-for-height = **thinness** and reflects **wasting**.
- ✚ Failure to gain sufficient weight relative to height or actual weight loss.

Generally resulted from:

- ✓ Acute (or short) periods of insufficient dietary intake.
- ✓ Repeated episodes of illness such as diarrhoea.
- ✚ Overweight: excess weight relative to height or insufficient height relative to weight.
- High WFH ($>+2$ Z-scores) is used to screen children at risk of developing obesity & future related morbidity such as heart disease.

Weight-for- Height or Length cont'd...

- Useful where age is uncertain.
- Independent of ethnicity: age 1-5years.
- In non-emergency situations, the highest prevalence of wasting generally occurs in young children (12–24 months of age).
- To identify whether a child is wasted or overweight, a child's actual weight is compared with that of a reference child of the same sex at exactly the same height.
- If prevalence of stunting is much higher than wasting in an area, use a combination of WFH & HFA.

Anthropometric indices: growth ...(1)

2. Length or Height-for-Age (L/HFA):

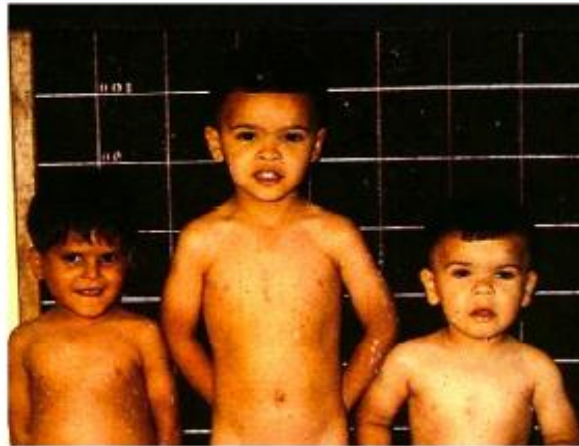
- An index of long term/ chronic/ past nutritional status.
- Low height-for-age relative to a reference = **shortness** and reflects **stunting**.
- Resulted from:
 - an extended period of inadequate food intake,
 - poor dietary quality & deficiencies of specific nutrients such as zinc & calcium,
 - repeated infections, or a combined effect.
- Low HFA commonly not appear **before 3 months of age** & prevalence generally increases with **age to 24–36 months**.

Length or Height-for-Age (L/HFA) cont'd...

- Stunting is associated with many negative outcomes:
 - ✓ Susceptibility to infections in childhood & increased mortality.
 - ✓ Decreased work productivity and lower incomes.
 - ✓ Low school performance (poor IQ, class repetition & dropouts).
- Application: HFA can be used to screen individual children for intervention, in regions where there is a known high prevalence of stunting.
- To identify whether a child is stunted, his/her actual height is compared with height of a reference child of the same sex at exactly the same age.

STUNTING (inadequate height-for-age)

125 cm
7 years old



103 cm
7 years old

100 cm
4 years old



Reference: Vilma Q. Tyler, UNICEF-CEE/CIS

Anthropometric indices: growth ...(2)

3. Weight-for-Age (WFA):

- Weight is influenced both by height and thinness.
- Low weight-for-age = **lightness** and reflects **underweight**.
- Low WFA (underweight) is a combination indicator of HFA & WFH.
- It is the most commonly reported & frequently used index.
- Used for monitoring growth & identifying children at risk of growth failure & malnutrition.
- Used in assessing the impact of intervention actions in growth monitoring programs.
- Has the highest predictive ability for childhood mortality.
- To identify underweight, a child's actual weight is compared with weight of a reference child of the same sex at exactly the same age.

Summary: WHO cut-offs for under nutrition in children

✦ The WHO recommends the use of the WHO child growth standards and cut-offs for under nutrition based on z-scores.

✦ Z-scores below **-2SD** are markers of undernutrition.

✦ Z-scores between **-2 and -3SD** are termed “moderate”

✦ While those **below -3SD** are termed “severe”

Indicator	Z -Score		
	<-3 -	-3 to -2	<-2
Weight-for-height	Severely wasted	Moderately wasted	Wasted
Height-for-age	Severely stunted	Moderately stunted	Stunted
Weight-for-age	Severely underweight	Moderately underweight	Underweight

Cut-off points for undernutrition based on percentage of the median for children 6-59 months

Nutrition Indicator	Moderate undernutrition	Severe undernutrition
Bilateral oedema	No	Yes
Weight for height (wasting)	$\geq 70\%$ to $<80\%$	$< 70\%$
Height for age (stunting)	$\geq 85\%$ to $<90\%$	$< 85\%$
Weight for age (underweight)	$\geq 60\%$ to $<80\%$	$< 60\%$

Anthropometric Indices: Growth ...(2)

4. Head Circumference-for-Age:

- Reflects **chronic protein energy malnutrition** in children <2 years.
- Abnormally low: intrauterine growth retardation or chronic malnutrition.
- Non nutritional factors: disease, genetic variation, cultural practices (binding head), difficult or forceps assisted delivery may affect head circumference.

Anthropometric indices...(3)

5. Body Mass Index (BMI): $\text{weight}/(\text{height})^2$

- Indicate body weight in relation to height, *i.e.* measure of overweight & obesity.
- High BMI (obesity level) is associated with type 2 diabetes & high risk of cardiovascular morbidity & mortality.
- Low BMI (underweight) reflects chronic energy deficiency.
- Height & weight measurements are quick, non invasive & more precise than skinfolds.

Advantages

- ✓ Easy, quick & non invasive.
- ✓ More precise than skin fold thickness.

Limitations

- ✓ Does not distinguish weight associated with muscle & fat: elevated BMI may be due to adiposity, muscularity, or edema.
- ✓ Do not indicate distribution of fat.
- ✓ Affected by ethnicity.

Classifying Nutritional status of Adults using BMI

overweight , obesity	Body mass Index (BMI (Kg/Ht in m ²))
Class III severe obesity	>40
Class I obese	30-34.9
Overweight	25- 29.9
Normal	18.5-24.9
Underweight	< 18.50,
Grade of CED (Chronic energy deficiency) (FAO); or Thinness (WHO).	
0	18.5Kg/m ²
I (Mildly Malnourished)	18.4- 17.0 Kg/m ²
II (Moderately Malnourished)	16.9 -16.0Kg/m ²
III (Severely Malnourished)	<16.0Kg/m ²

BMI <16.0Kg/m² is associated with :

- LBW in 50% of cases
- Decreased work capacity
- Poor resistance to infection

Classifying Nutritional status of 5-19 years using BMI

Age 5-19 years	Nutritional Status
<-3 SD	Severely Malnourished
-2 to -3 SD	Moderately malnourished
-1 to -2 SD	Mildly malnourished
>-1 SD	Normal weight

Meanings of the Indices Derived from Growth Measurements

Weight for Age = $\frac{\text{Weight of the child}}{\text{Weight the normal child of the same age}} \times 100$

Weight for height = $\frac{\text{Weight of the child}}{\text{Weight of the normal child of the same height}} \times 100$

Height for age = $\frac{\text{Height of the child}}{\text{Height of the normal child of the same age}} \times 100$

Anthropometric Indicators: an index plus a cut-off point. Example:

- ❑ W F A < 60% = is indicator of severe underweight.
- ❑ H F A < 85% = indicator of severe stunting.
- ❑ W F H < 70% = is indicator of severe wasting.

2. Anthropometric Assessment of Body Composition

- Mostly used in clinics & hospital settings to identify under or over nutrition & evaluate effectiveness of nutrition intervention.
- Body composition methods measure the amount and distribution of fat and lean body mass (protein or muscle & bone).
- In some of these methods, it is assumed that the body is a two-compartment model of fat and fat-free mass (lean mass).
- The three-compartment model assumes the body is composed of fat mass, lean mass and bone.
- The four-compartment model divides the body into water, lean mass, mineral (bone) and fat.

2.1 Assessment of Body fat

Skin fold Thickness: estimates size of subcutaneous fat.

- No agreement on best site as index of total body fat.
- Four sites that are generally suggested for skinfold measurements include the triceps, biceps, subscapular, and supra iliac skinfold sites.
- **Triceps skinfold thickness:** estimate size of subcutaneous fat deposit.
- Skinfold measurement is inexpensive and has no risk to the patient.

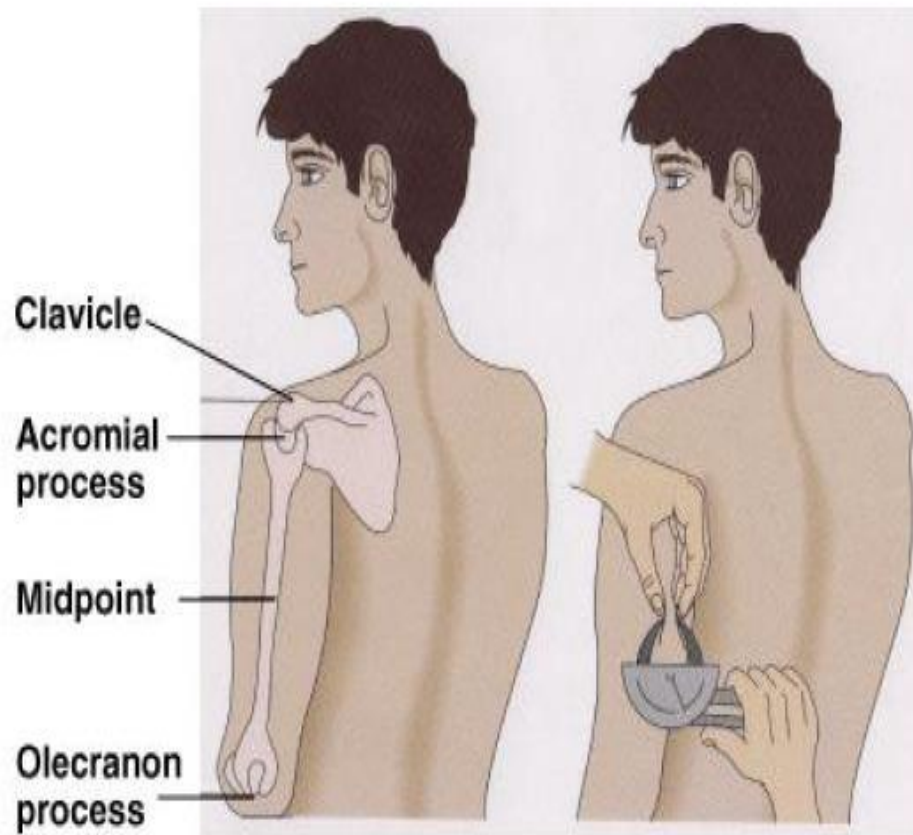
Assessment of body fat... (1)

Multiple skinfolds:

- For total body fat, one limb skinfold (*e.g.*, triceps) plus one body skinfold (subscapular) is recommended.
- Distribution of body fat depends on ethnicity, age & sex.
- Skinfold measurement is of considerable value in assessing the amount of fat & therefore the **reserve of energy in the body**.

Measurement of Triceps skinfold thickness

Skin fold measurement



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Location of Sub-scapular (A) & Suprailiac (B) skinfold sites

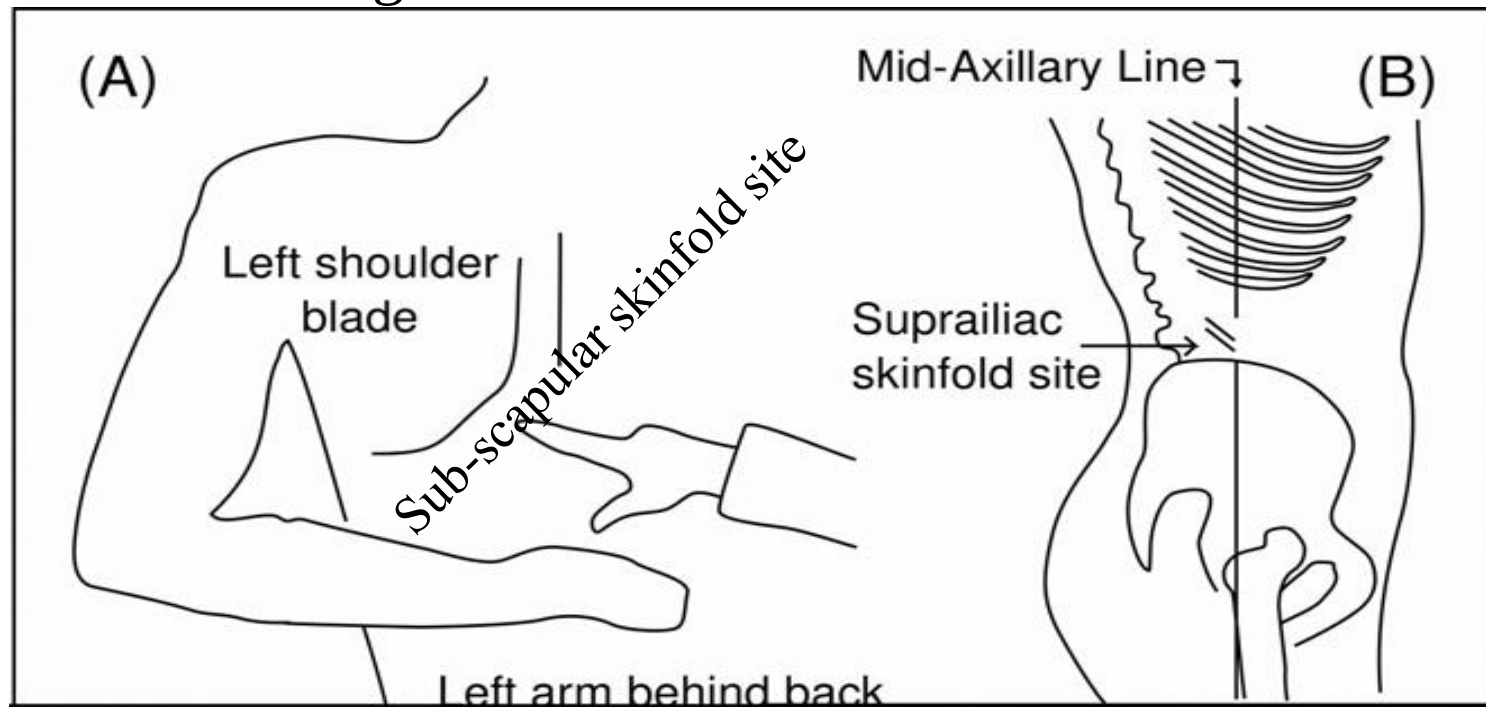
(a) Below & laterally to angle of shoulder blade.

- ✓ Placing arm behind back helps locate the site.
- ✓ Skinfold = 45° from horizontal.



(b) Mid - axillary line superior to iliac crest.

- ✓ Skinfold picked up obliquely posterior to midaxillary line, parallel to cleavage lines of skin.



Assessment of body fat... (2)

B. Waist Circumference (WC): proxy for central fat.

- Waist circumference is measured at the level of the umbilicus to the nearest 0.5 cm.
- It predicts mortality better than any other anthropometric measurement.
- WC measurement alone can be used to assess obesity, and two levels of risk have been identified.

	MALES	FEMALES
LEVEL1	> 94cm	> 80cm
LEVEL2	> 102cm	> 88cm

- Level 1 is the maximum acceptable WC level (if there is no further weight gain).
- Level 2 denotes obesity and requires weight management to reduce the risk of type 2 diabetes & CVD's complications.

Waist Circumference Measurement

- The subject stands erect with relaxed abdominal muscles, arms at the side, and feet together.
- No universally accepted method for WC measurement.
 - ✓ Taken around natural waist at its widest point: WHO (Fig. A).
 - ✓ Midway between the lowest ribcage & anterior superior iliac spine OR at least 2.5cm above belt (Fig. B).

A



B



Waist circumference measurement



Hip Circumference

- Is measured at the point of greatest circumference around hips & buttocks to the nearest 0.5 cm.
- The subject should be standing and the measurer should squat beside him.
- Both measurement should taken with a flexible, non-stretchable tape in close contact with the skin.

Hip circumference measurement



Waist-hip circumference Ratio (WHR)

- Distinguishes fatness in lower trunk (hip & buttocks) (mainly subcutaneous) & upper trunk (waist & abdomen).
- WHR correlates with total body fat.
- Elevated WHR strongly associated with increased risk of Coronary Heart diseases (CHD), stroke, type 2 diabetes.

Interpretation of WHR

- If the WHR is **>1** in male (indicates central/ upper body obesity), & **>0.87** in female there is high risk of CHD, stroke, type 2 diabetes.
- A WHR below these cut-off levels is considered as low risk.

Table 2a: Classification of Overweight and Obesity by BMI, Waist Circumference, and Associated Disease Risk*

		<u>Normal:</u>		<u>Obese:</u>
Waist Circumference		Men:	≤ 40 in (≤ 102 cm)	> 40 in (> 102 cm)
		Women:	≤ 35 in (≤ 88 cm)	> 35 in (> 88 cm)
BMI (kg/m²)	Obesity Class		↑Disease Risk* Relative to Normal Weight and Waist Circumference)	
Underweight	< 18.5		–	–
Normal**	18.5-24.9		–	–
Overweight	25.0-29.9		Increased	High
Obesity	30.0-34.9	I	High	Very High
	35.0-39.9	II	Very High	Very High
Extreme Obesity	≥ 40	III	Extremely High	Extremely High

*Disease risk for Type 2 Diabetes Mellitus, Hypertension, and CVD.

**Increased waist circumference can also be a marker for increased risk even in persons of normal weight.

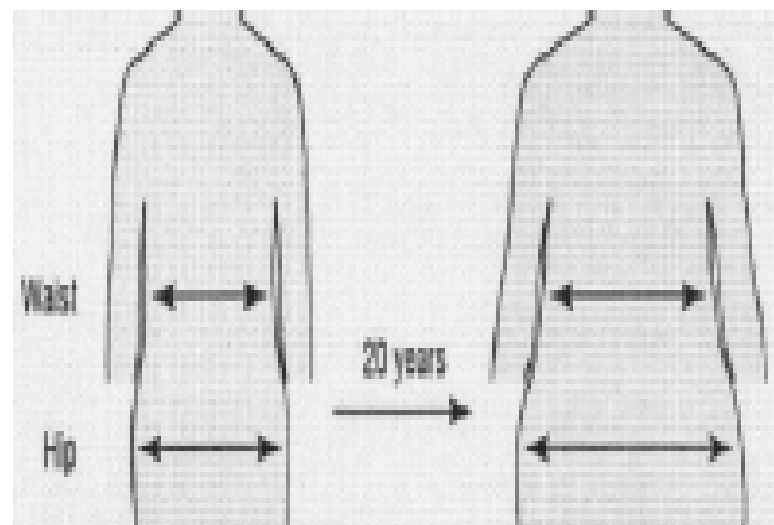
Adapted from “Preventing and Managing the Global Epidemic of Obesity. Report of the World Health Organization Consultation of Obesity.” WHO, Geneva, June, 1997.

Limitations of Waist: Hip circumference ratio

- W:H ratio measurement can be misleading in longitudinal studies.

- Here W:H ratio is stable over 20 yrs despite accumulation of visceral adipose tissue which would have been predicted from 20cm increase in waist

- Thus waist circumference is a crude index of the absolute amount of abdominal tissue whereas W:H ratio is an index of the relative accumulation of abdominal fat



BMI	24	35
Waist	80cm	100cm
Hip	100cm	125cm
WH ratio	0.8	0.8

From BMJ March (2001)

High Technology methods for assessing body composition

- Bone mineral density by Dual energy X-ray absorptiometry (DEXA):
- Underwater weighing using Archimedes principle
- Bioelectrical Impedance (BIA)
- Total body water by Isotope dilution
- Total body nitrogen by in vitro neutron activation analysis
- Total Body Potassium
- Ultrasound
- MRI computed & tomography
- Air Displacement Plethysmography (ADP or Bod Pod)

#Assignment 2. Write a short note on the application of the above High Technology methods .

2.2 Assessment of Fat free mass

- Fat free mass is a mixture of water, protein, and minerals, with muscle serving as the major protein store.
- Mid-upper arm circumference (MUAC) is widely used method to assess fat free mass.
- Provides an index of the protein reserves of the body.
- Decrease in MUAC may reflect loss in muscle mass or subcutaneous fat or both.
- Where the amount of fat is small, MUAC changes parallel to changes in muscle mass.

2.2.1 Mid-Upper Arm Circumference (MUAC)

- MUAC is a proxy indicator of nutrient reserves and wasting.
- Measurement is not time consuming, and effective predictor of risk of death for under 5 years of age.
- Endorsed as an independent admission criterion for nutrition programmes addressing severe acute malnutrition (SAM).
- It is also used for screening.
- The cut-off point for SAM is **< 11.5 cm**.

Mid-upper arm circumference cont ...

- Subject should stand erect sideways to the measurer and with head in Frankfurt plane.
- Measurement taken at the mid-point of upper left arm using flexible & non-stretch tape.
- Precision of measurement must be high as MUAC varies little at any given age.
- Difference between trainee & trainer should be 0 to 5 mm.



Classifying Nutritional status of children 6 months to 5 years old using MUAC

MUAC cut-off points	Nutritional Status
< 11.5 cm	Severe Acute Malnutrition (SAM)
≥11.5 - <12.5 cm	Moderate Acute Malnutrition (MAM)
≥12.5 cm	Not malnourished (Normal)

MUAC cutoff for adult pregnant and lactating women

	Nutritional Status		
	Severe acute malnutrition	Moderate acute malnutrition	Normal
MUAC level	<18.0cm	18-21 cm	> 21 cm

Classifying Nutritional status of 5-14 years using MUAC

Age in years	MUAC level	Nutritional Status
5-9	<135 mm	Severe malnutrition
10-14	<160 mm	
5- 9	≥ 135 mm - <145 mm	Moderate malnutrition
10-14	≥ 160 mm - <180 mm	
5- 9	≥ 145 mm	Normal
10-14	≥ 180 mm	

Evaluation of measurement error in Anthropometric measurements

Measurement	Trainee-trainer difference		
	Good	Fair	Poor
Height or length (mm)	0 – 5	6 – 9	10 – 19
Weight (kg)	0 – 0.1	0 – 2	0.3 – 0.4
Arm circumference (mm)	0 – 5	6 – 9	10 – 19
Skinfolds (any) (mm)	0 – 0.9	1.0 – 1.9	2.0 – 4.9

After Zerfas (1985). Differences greater than those under
“Poor” should be taken to indicate gross error

Bilateral Pitting Oedema

- Bilateral pitting oedema can be verified when thumb pressure is applied on top of both feet for three seconds, and then leaves a pit (indentation) which remains for several seconds on both feet after the thumb is lifted.
- It usually starts in the feet and ankles.
- It is important to test both feet; if the pitting is not bilateral, the oedema is not of nutrition origin.
- There are three grades of bilateral pitting oedema.
- When there is no bilateral pitting oedema, the grade is absent.
- Grades of bilateral pitting oedema are classified by “+” signs.

Bilateral Pitting Oedema

Grade+

Mild: both feet/ankles



Grade++

Moderate: both feet, plus lower legs, hands, or lower arms



Grade+++

Severe: generalized oedema including both feet, legs, hands, arms and face



Nutrition Indicator

Moderate Acute Malnutrition (MAM)

Severe Acute Malnutrition (SAM)

WFH (wasting)

≥ -3 SD & < -2 SD

< -3 SD

MUAC

≥ 115 mm & < 125 mm
(≥ 11.5 cm & < 12.5 cm)

< 115 mm
(< 11.5 cm)

MUAC for age/height

< -3 SD

Bilateral Oedema

No

Yes

Anthropometric Reference Data

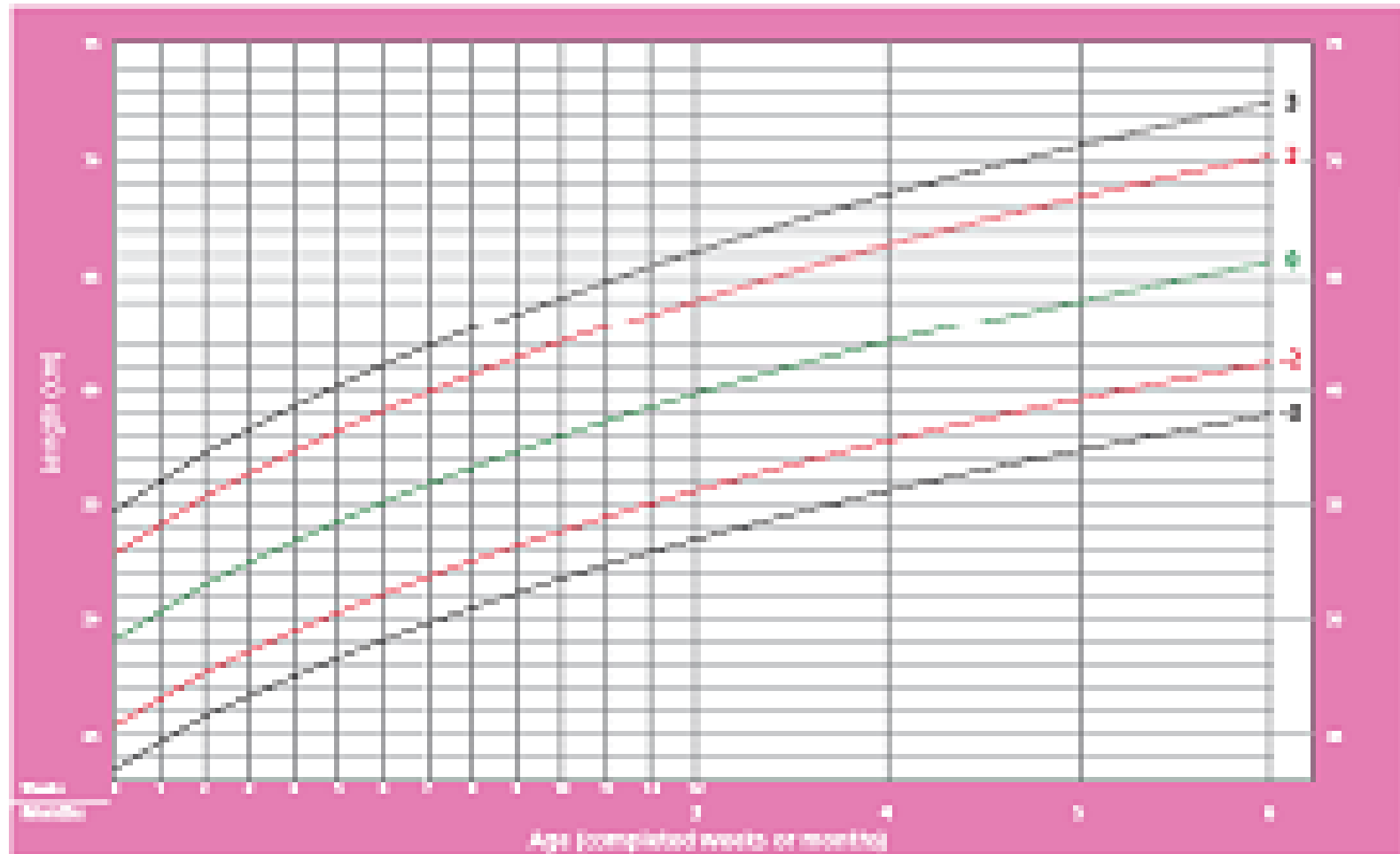
International growth reference data

- UN National Center for Health Statistics (NCHS)/ WHO 1977 reference data for children & adolescents 2 to 18 years.
- NCHS/ WHO (1983) recommended for children >5 years.
- Center for Disease Control & Prevention (CDC) 2000 growth data is also available.
- WHO 2005 growth curves: for children < 5 years.
- WHO 2007 reference is a reconstruction of the 1977 (NCHS)/ WHO reference: for children <5years.
- For adults: Age-specific BMI percentile cutoffs.

New WHO growth standard: for girls (Z-scores)

Length-for-age GIRLS

Birth to 6 months (z-scores)

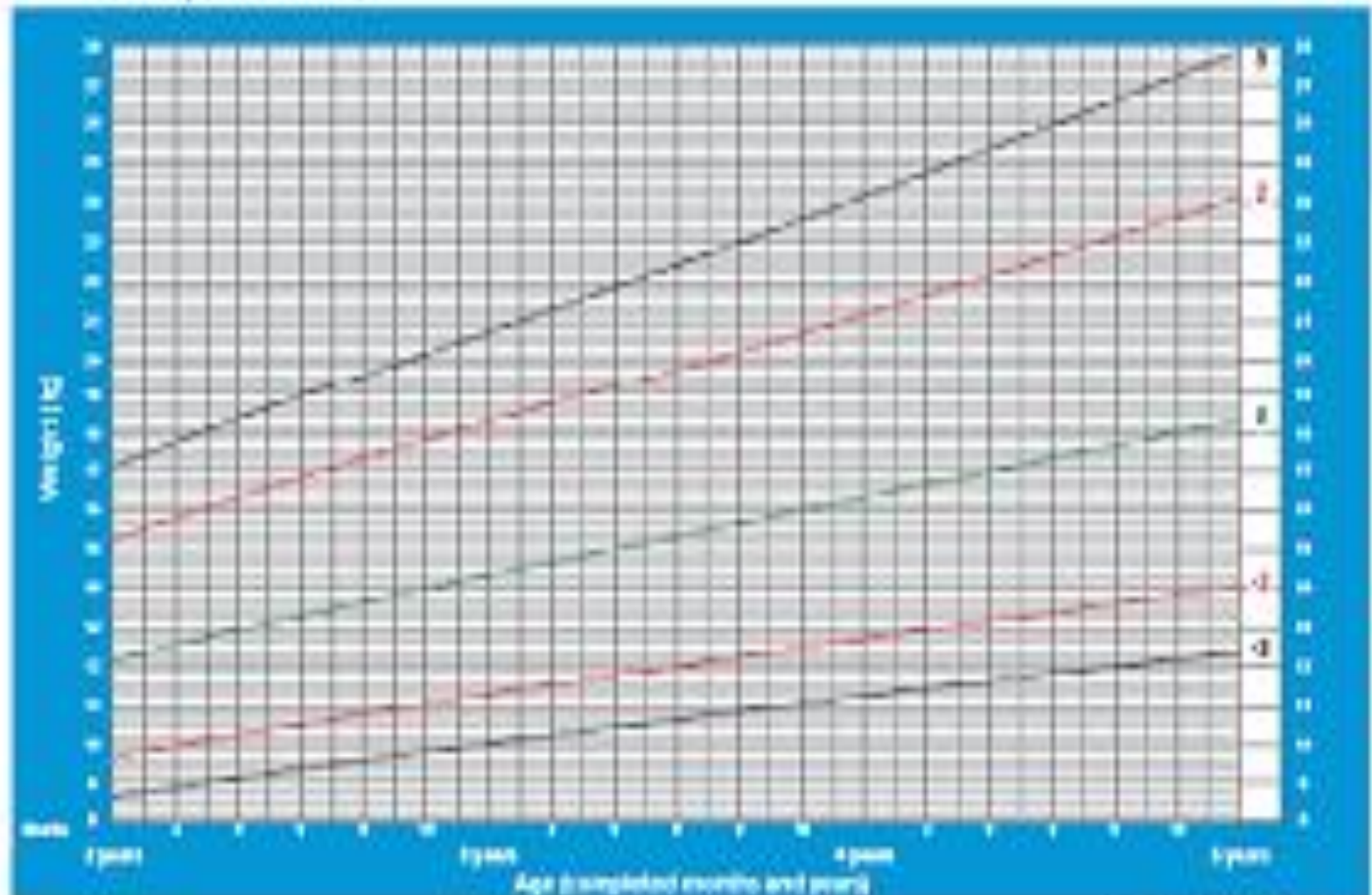
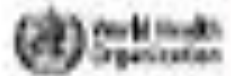


WHO-GIRL GROWTH STANDARD

New WHO growth standard: for boys (Z-scores)

Weight-for-age BOYS

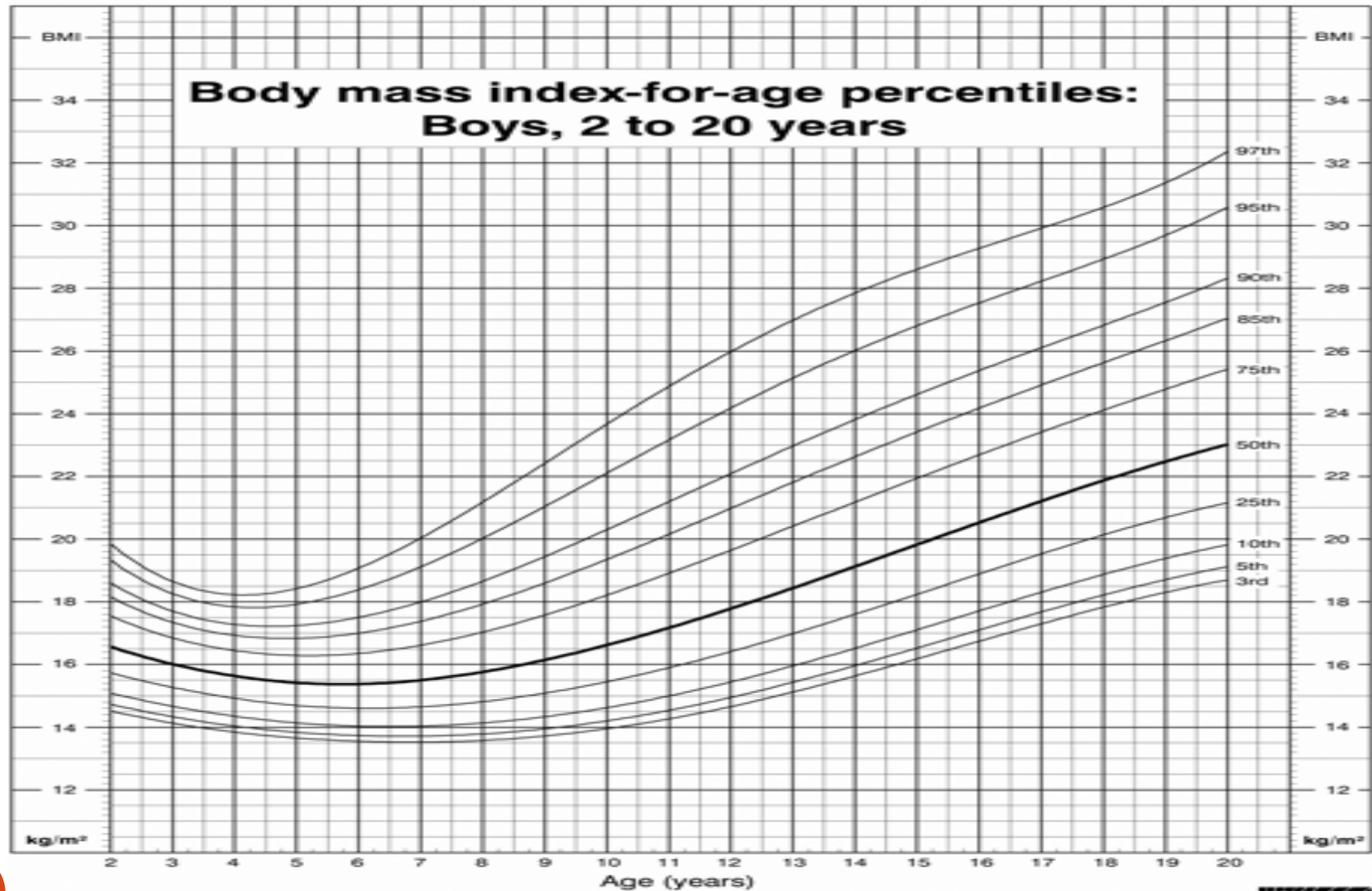
2 to 5 years (Z-scores)



WHO Child Growth Standards

CDC Growth Charts – BMI for age, Boys 2 to 20 y

CDC Growth Charts: United States



Evaluation of Anthropometric Indices

A. How to compare growth indices with reference data?

- Taking age and sex into consideration, differences in measurements can be expressed in a number of ways:
 - i. Z-scores: Used for developing countries
 - ii. Percent of the median
 - iii. Percentiles: Used for developed countries

The Z-score or standard deviation (SD) score

- Is the difference b/n the value for an individual & the median value of the reference population for the same age or height, divided by the SD of the reference population.
- The most commonly used cut-off with Z-scores is -2SD, irrespective of the indicator used.
- This means children with a Z-score for underweight, stunting or wasting, below -2SD are considered moderately or severely malnourished.
- Used for low income countries.

$$\text{Z-score or (SD score)} = \frac{\text{Observed value} - \text{Median reference value}}{\text{SD of reference population}}$$

Example

Child	Sex	Age	Weight (kg)	Height (cm)	Median ht	Median wt	SD
1	Male	6 yr 5 mo	13.3	107.5	118.5	W/A = 21.6 W/H = 17.9	H/A = 4.9 W/A = 3.2 W/H = 1.6

- $$\text{H/A Z score} = [(\text{observed height} - \text{median reference height}) / \text{standard deviation (SD)}]$$

$$= (107.5 - 118.5) / 4.9 = \underline{-2.25 \text{ Z}} \text{ (Stunted)}.$$
- $$\text{W/A Z score} = [(\text{observed weight} - \text{median reference weight}) / \text{SD}]$$

$$= (13.3 - 21.6) / 3.2 = \underline{-2.59 \text{ Z}} \text{ (Underweight)}.$$
- $$\text{W/H Z score} = [(\text{observed weight} - \text{median reference weight}) / \text{SD}]$$

$$= (13.3 - 17.9) / 1.6 = \underline{-2.86 \text{ Z}} \text{ (Wasted)}.$$

Use of distance growth charts: eg. WHO (2006)

- Plot a single measurement (e.g., length) on a reference distance growth chart.
- Read off Z-score range within which value falls from chart.
- Assess whether subject is at risk of malnutrition (e.g., less the - 2.0 Z- scores on reference data)



Percentage of the Median

- The median is the value at **exactly the midpoint between the largest & smallest value.**
- The ratio of a measured or observed value in the individual to the median value of the reference data for the same age or height for the specific sex, expressed as a percentage.
- If a child's measurement is exactly the same as the median of the reference population, we say that they are “100% of the median”.

$$\text{Percent of median} = \frac{\text{observed value (Weight or height)}}{\text{median value of reference population (Weight or Height) of the same age}} \times 100$$

Cut off points for Acute malnutrition (Weight-for-Height)

- Acute malnutrition based on weight-for-height in z-scores and percentage of the median.

	Degree of malnutrition	Definition using z-score	Definitions using % of median
Acute	None/Mild	≥ -2.0	$\geq 80\%$
	Moderate	≥ -3.0 but < -2.0	$\geq 70\%$ but $< 80\%$
	Severe	< -3.0 or oedema	$< 70\%$ or Oedema
Global Acute (GAM)	Moderate + Severe	< -2.0 and/or Oedema	$< 80\%$ and/or Oedema
Severe Acute (SAM)	Severe	< -3.0 and/ or Oedema	$< 70\%$ and/or Oedema

Cut off points for Chronic malnutrition (Height-for-Age)

- Chronic malnutrition based on Height-for-Age in z-scores and percentage of the median.

	Height for age z-scores	Height for age % of median
Normal/Not Stunted	≥ -2 z-scores	$\geq 90\%$
Moderate chronic malnutrition	≥ -3.0 but < -2.0	$\geq 80\%$ and $< 90\%$
Severe chronic malnutrition/Severely stunted	< -3 Z scores	$< 80\%$
Total chronic malnutrition/Total stunted (moderate + severe)	< -2 Z score	$< 90\%$

Cut off points for Underweight (Weight-for-Age)

- Underweight based on Weight-for-Age in Z-scores and percentage of the median.

Description of Nutritional Status	Weight for Age Index Z scores	Weight for Age % of median
Severe Underweight	<-3 Z scores	<70%
Moderately Underweight	≥ -3.0 but <-2.0	$\geq 70\%$ and <80%
Total Underweight (moderate plus severe)	<-2 Z score	<80%
Normal	≥ -2 Z-scores	$\geq 80\%$

Percentiles

- The percentile is the rank position of an individual on a given reference distribution.
- Percentiles are stated in terms of what percentage of the group that the individual equals or exceeds.
- The percentile reference limits commonly used for designating individuals “at risk” to malnutrition in reference to NCHS centiles either below 3rd or 5th percentile or above 97th or 95th percentiles.
- Usually the 3rd centile is taken as cutoff point for labeling malnourished individual.

Percentiles cont'd...

- *E.g.*, the child's height-for-age being at the 80th percentile means that 80% of children of the same age in the reference population have a height lower than his height (are shorter than him).
- If a boy is at the 10th percentile with a height of 120.5 cm, it means 10% of boys have a height below 120.5 cm.
- Used for affluent countries.
- Not recommended for low income countries because some persons may have indices *below extreme percentiles*.

Indices and cut-off points for defining malnutrition

Measurements	Indices	Indicators	Cut-off points for the deferent ways defining Malnutrition		
			Centile	Percent of the reference median	Z score Or SD
Weight and Height	Weight for Height(WFH)	Wasting	<3 rd P	<80	<-2
Height and Age	Height for Age(HFA)	Stunting	<3 rd P	<90	<-2
Weight and Age	Weight for Age(WFA)	Underweight	<3 rd P	<80	<-2
Weight, Height and Age	BMI for age	Thinness	-	-	<-2

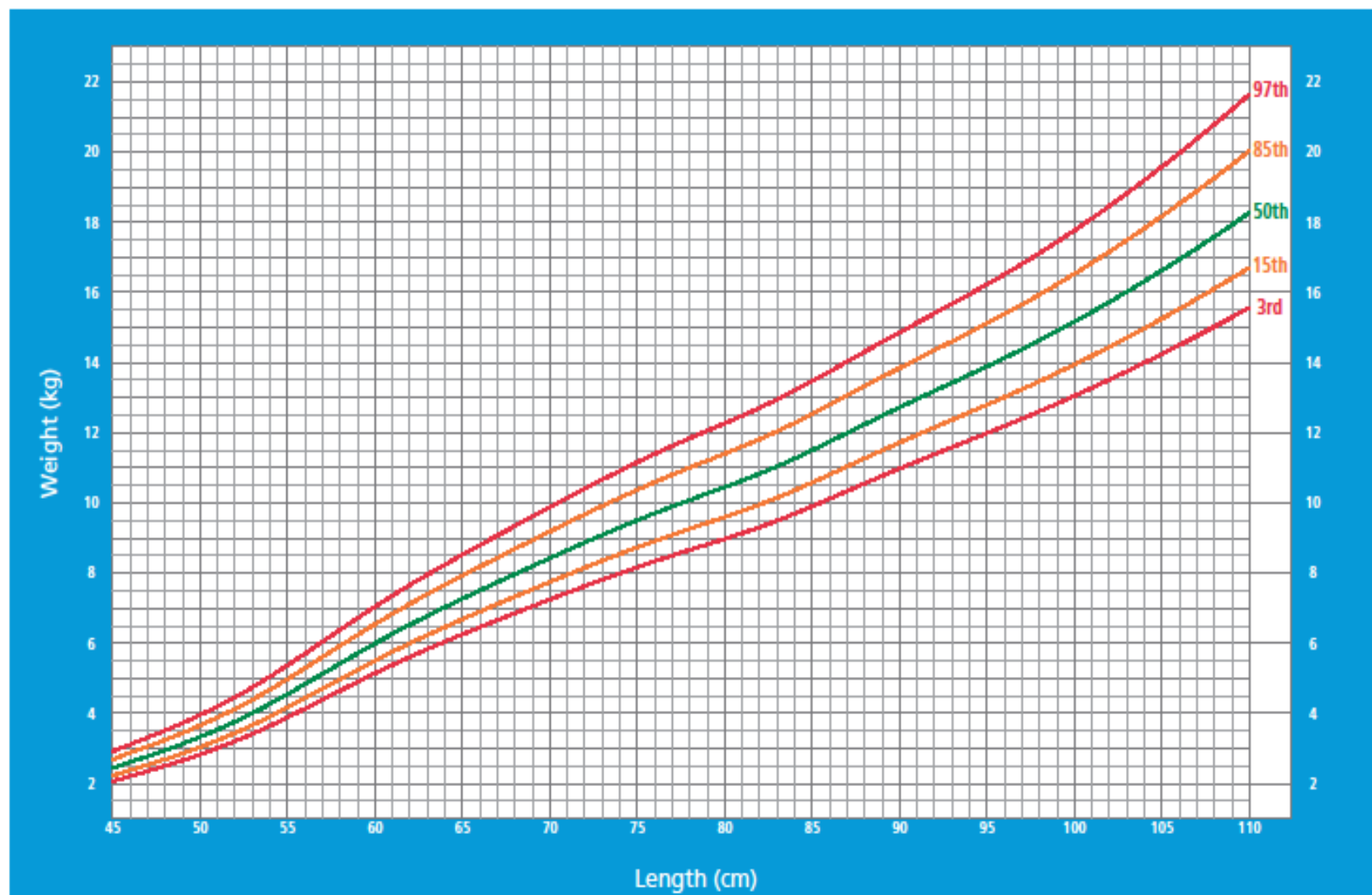
“Moderate” malnutrition classified as the percent falling between ≤ -2 to ≥ -3 SD and “severe” as the percent falls below -3 SD from the reference median.

Exercise

- Plot the weight & length percentiles of a 12 month old boy weighing 10 k.g and being 75 cm.
- What is his weight & length percentile?
- How do you interpret it?

Weight-for-length BOYS

Birth to 2 years (percentiles)



WHO criteria for assessing *severity* of under nutrition in children < 60 months

Indicator	Percent Prevalence			
	Low	Medium	High	V High
Underweight (Lightness)	<10	10-19.9	20-29.9	<u>>30</u>
Stunting (Shortness)	<20	20-29.9	30-39.9	<u>≥ 40</u>
Wasting (Thinness)	<5	5-9.9	10-14.9	<u>≥15</u>

From Gorstein et al. (1994)

Growth Monitoring and Promotion (GMP)

- GMP is a periodic measurement of the weight of under five children.
- It is important to detect nutritional problems earlier.
- GMP is a simple and powerful tool to:
 - Monitor the growth of children.
 - Detect growth faltering early (before it becomes too late!).
 - Assess and make the growth pattern of a child visible to the parents.

Using a child growth card to assess nutritional status (Weight for Age)

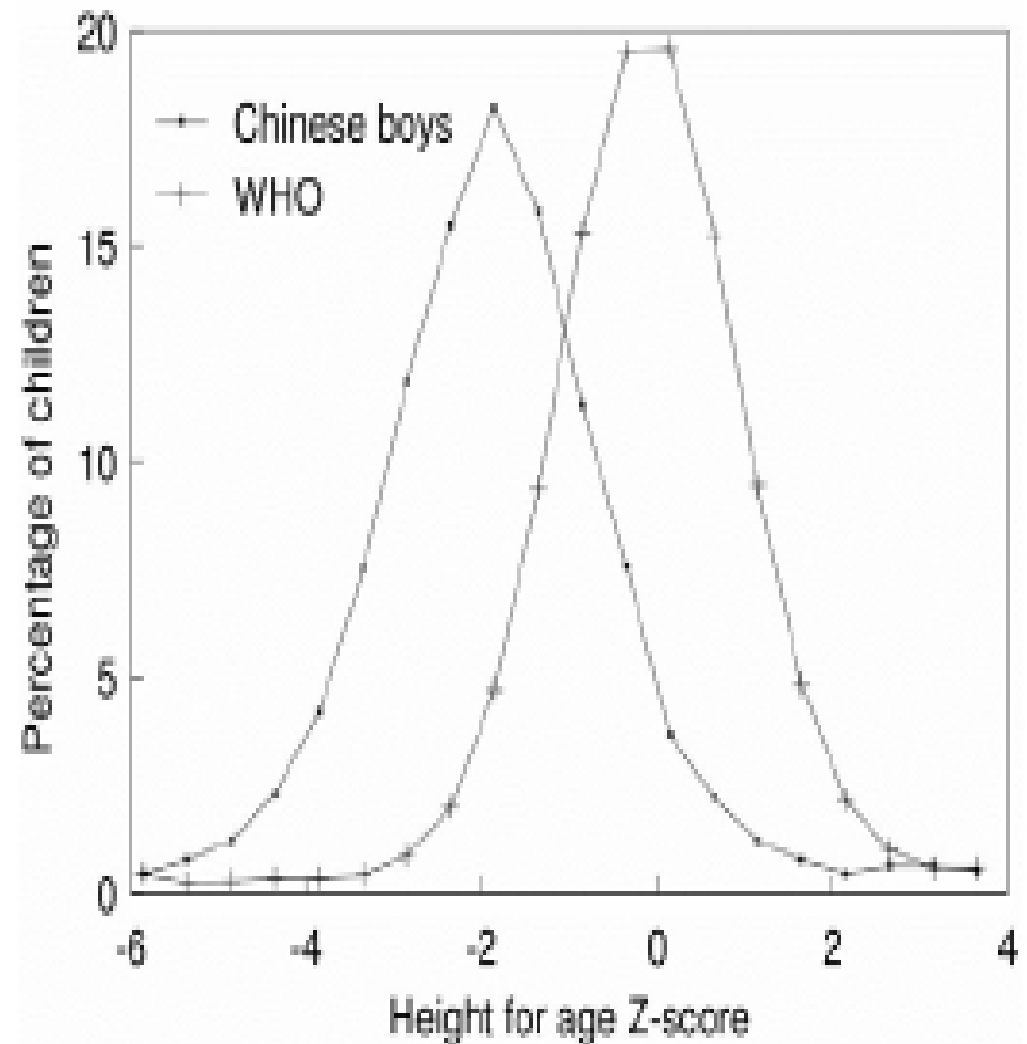
- Determine age of child in months.
- Weigh child (lightly dressed).
- Plot weight of child against age in months.
- Determine if point is normal or indicating undernutrition.
- For screening and/or assessing response to interventions multiple growth indices will be used, (*i.e.* weight-for-height; height-for-age, weight-for-age).
 - Each can be classified as: low, normal & high based on reference limits: percentiles or Z-scores.

Uses of Anthropometric data: At population level

- For international comparisons of growth.
- To identify trends in growth over time.
- For nutritional surveillance: continuous monitoring of population in emergencies.
- For targeting interventions & assessing response to interventions.
- Identifying determinants and consequences of malnutrition.

Comparison of Z-score distribution for height-for age of Chinese boys with reference (NCHS/WHO)

- Shows a high prevalence of stunting (i.e., ht-for-age < -2 SD) (~18%) in these Chinese boys compared to reference
- Can also be used as a screening tool to identify groups at high risk
- Appropriate interventions can then be designed and targeted to reduce stunting:
 - improve diet quality
 - improve hygiene and potable water supplies
 - prevent and treat infectious diseases



BMI and measures of body fat and Disease risk

- BMI said to be an independent index of body fat.
 - ✓ Subjects with same BMI said to have *same relative fatness* irrespective of age, sex & ethnicity.
- BUT: relationship b/n BMI & body fat depends on:
 - ✓ **Age:** older women more percent fat than younger women with same BMI.
 - ✓ **Sex:** women more fat than men for same BMI.
 - ✓ **Ethnicity:** Chinese, Asian Indians & Indonesian have more percent fat than Caucasians for same BMI and Pacific islanders have less percent fat for same BMI.

WHO's classification to assess public health significance of low BMI (<18.5 k.g/m²)

Situation	Percent with BMI (<18.5 k.g/m ²)
Low prevalence	5-9
Medium prevalence	10-19
High prevalence	20-39
Very high prevalence	>40

WHO classification scheme to screen for overweight for adults based on BMI

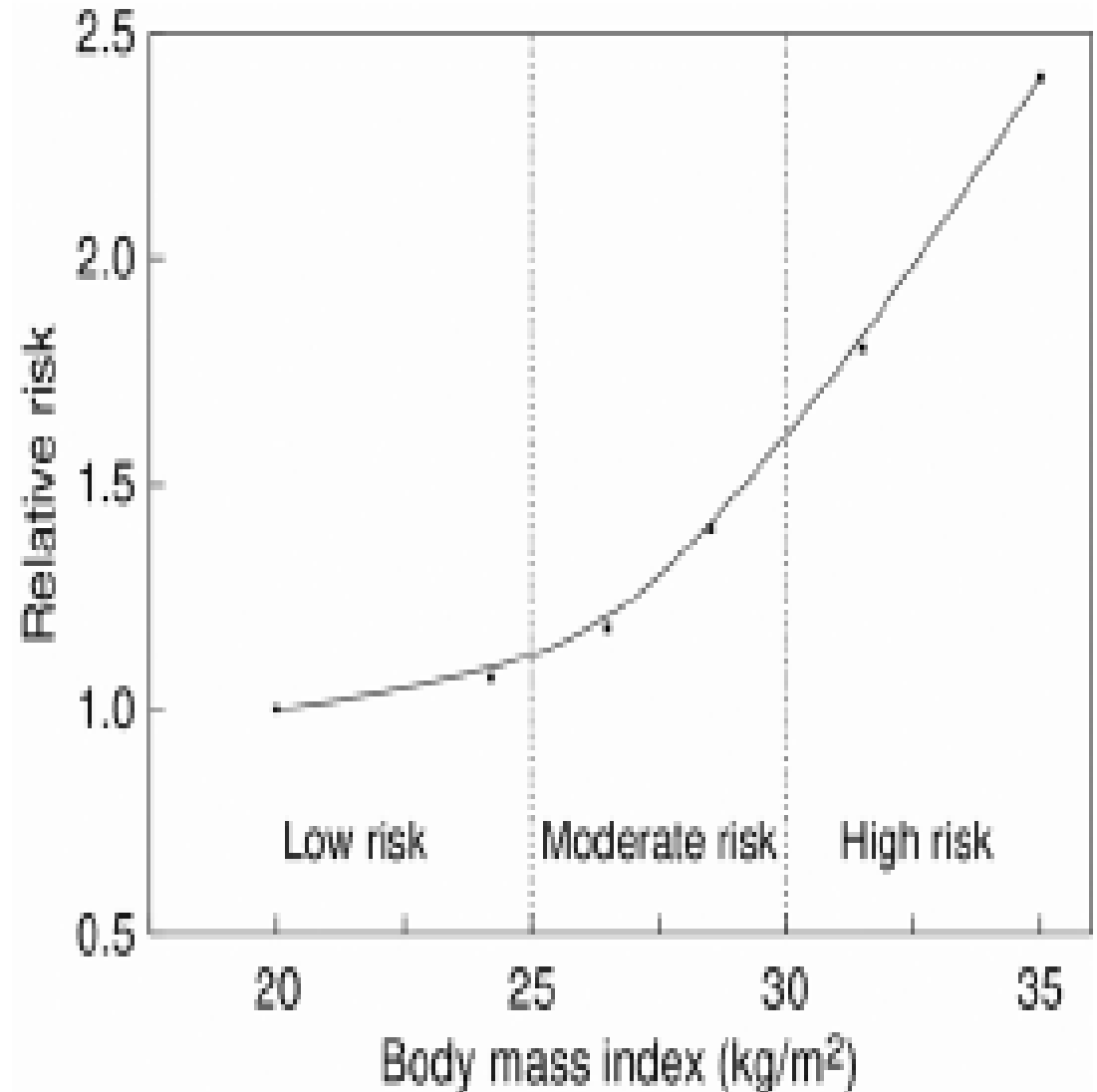
Classification	BMI	Risk of comorbidities
Normal range	18.5 - 24.99	Average
Overweight	≥ 25	
Pre-obese	25.0 - 29.99	Increased
Obese class I	30.0 - 34.99	Moderate
Obese class II	35.0 - 39.99	Severe
Obese class III	≥ 40.00	Very severe

Relationship between BMI & relative risk of mortality

As BMI increases, risk increases of:

- High blood pressure
- Type 2 diabetes mellitus
- Other CVD risk factors
- Mortality

From WHO (2000)



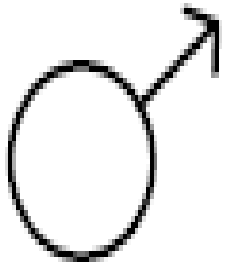
Health risks also associated with regional body fat distribution as well as total body fat

Fat can be stored primarily in the central or peripheral regions

Central

OR

Peripheral



android

gynoid

upper body

lower body

apple



pear



Vs

Health risks associated with central body fat

Central fat consists of:

- ✓ Visceral (intra-abdominal) fat and;
- ✓ Subcutaneous fat.
- Increased health risk mostly due to **visceral fat**.
- **Central fat** tends to be greater in men; related to testosterone and other factors and it increases with age.
- Women will have more central fat after menopause.
 - Protected by oestrogen in pre-menopausal women.

Methods for screening for overweight in Children and Adolescents

- **Weight-for-height** : $>+2$ Z-scores for children.
- **BMI percentiles**: 85th percentile: risk of overweight; & $> 95^{\text{th}}$ percentile: overweight (U.S classification).
- Boys with BMI $>75^{\text{th}}$ percentile have increased risk of mortality from all causes of CHD, Cerebrovascular disease in later ages.

Advantages of Anthropometric Measurements

- ☑ Use simple, safe, non-invasive techniques.
- ☑ Equipment used is non-expensive, portable & durable & can be made or purchased locally.
- ☑ Unskilled personnel can perform measurement procedures with minimal training.
- ☑ Methods are precise, and accurate provided that standardized techniques are used.
- ☑ Information is generated on past long term nutritional history which cannot be obtained with equal confidence using other techniques.
- ☑ They can also supply information on malnutrition to families and health care workers prior to the onset of severe growth failure.
- ☑ The procedures can assist in the identification of mild to moderate malnutrition as well as severe states.
- ☑ The methods may be used to evaluate changes in nutritional status overtime and the secular trend.

Limitations of Anthropometry

- ❑ **Inter-observers errors** in measurements (*i.e.* subjected to both random & systematic measurement errors).
- ❑ Problems with **reference standards** *i.e.* **local versus international standards**.
- ❑ Relatively **insensitive** methods and **cannot detect or identify specific nutrient deficiencies**.
- ❑ Need for **precise age data** in young children for construction of most indices.
- ❑ Unable to distinguish **disturbances** between growth or body composition induced by **nutrient deficiencies** (**protein or energy**).
- ❑ Certain **non-nutritional factors** (**disease state, reduced energy expenditures** etc.) can reduce the specificity & sensitivity.
- ❑ **Debate over selection of appropriate reference data and cut-off points** to determine conditions of abnormality.

Conclusions/ Key messages

1. Anthropometry is the use of body measurements to assess & classify nutritional status in an individual.
2. Body measurement includes weight, height/length & MUAC for children > 6 months of age.
3. Visible wasting & bilateral oedema are clinical signs of acute malnutrition.
4. Acute malnutrition among infants <6 months of age is assessed using visible sign of wasting and bilateral oedema. Social criteria such as an absent mother & inadequacy of breastfeeding can indicate nutritional risk.
5. Acute malnutrition among children 6-60 months of age is assessed by using weight-for-height/ length, MUAC & bilateral oedema.
6. Under nutrition among children & adolescents (5-19 years of age) is assessed by Body Mass Index-for-age & clinical signs.

Conclusions/ Key messages

7. Adult under nutrition is assessed by using BMI, MUAC & clinical signs. MUAC is the preferred nutritional index during pregnancy & up to 6 months postpartum.
8. Even though there are numerous issues related to the assessment of under nutrition in elderly, BMI is recommended, in addition to clinical signs.
9. The use of 2006 WHO growth standards is now recommended over the use of 1978 NCHS growth references in the definition of acute malnutrition in children 6-60 months. The 2007 WHO growth reference is recommended for use in assessment of children & adolescents 5-19 years.
10. Nutritional indices should be presented as Z-scores as opposed to percentage of the median (percentage of the median is no longer recommended for use in classification of individuals nutritional status in developing countries).

End of Chapter 3

Thank you!